

Variation of Parameters



The Wronskian

- 1 For $y_1 = e^x$, $y_2 = e^{2x}$ (solutions of $y'' - 3y' + 2y = 0$), compute the Wronskian $W = y_1 y_2' - y_1' y_2$.
 - 2 For $y_1 = \cos x$, $y_2 = \sin x$, compute W .
-

The variation of parameters formula

- 3 State the variation of parameters formula for a particular solution of $y'' + p(x)y' + q(x)y = g(x)$, given independent solutions y_1, y_2 of the homogeneous equation with Wronskian W .
-

A forcing term with no undetermined-coefficients trial form

- 4 Solve $y'' + y = \tan x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$ using variation of parameters.
-

A repeated root, with forcing dividing by x

- 5 Solve $y'' - 2y' + y = \frac{e^x}{x}$ ($x > 0$) using variation of parameters.
-

Comparing with undetermined coefficients

- 6 Solve $y'' - y = e^{2x}$ by variation of parameters, then check the particular solution using undetermined coefficients.
-

The Wronskian for complex-root solutions

- 7 For $y_1 = e^{-x} \cos 2x$, $y_2 = e^{-x} \sin 2x$, compute the Wronskian $W = y_1 y_2' - y_1' y_2$.
-

Setting up variation of parameters (without full integration)

- 8 For $y'' + 4y = \sec 2x \tan 2x$, with $y_1 = \cos 2x$, $y_2 = \sin 2x$, $W = 2$, set up the two integrals in the variation-of-parameters formula (don't evaluate).
-

A polynomial forcing term, cross-checked against undetermined coefficients

- 9 Solve $y'' + y = x$ using variation of parameters ($y_1 = \cos x$, $y_2 = \sin x$, $W = 1$), and verify it matches the undetermined-coefficients answer $y_p = x$.
-

A repeated root with a forcing term dividing by x-squared

- 10 Solve $y'' + 2y' + y = \frac{e^{-x}}{x^2}$ ($x > 0$) using variation of parameters.
-

Summary: choosing undetermined coefficients vs. variation of parameters

- 11 For $y'' - y = e^{2x} \tan x$, explain why variation of parameters is required rather than undetermined coefficients.
-

One more Wronskian computation

- 12 For $y_1 = e^{3x}$, $y_2 = e^{-x}$ (solutions of $y'' - 2y' - 3y = 0$), compute the Wronskian $W = y_1 y_2' - y_1' y_2$.
-

A trigonometric forcing term via variation of parameters

- 13 Solve $y'' + y = \sec x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$ using variation of parameters ($y_1 = \cos x$, $y_2 = \sin x$, $W = 1$).
-

One final setup exercise

- 14 For $y'' - 4y' + 4y = \frac{e^{2x}}{x^3}$ ($x > 0$), with $y_1 = e^{2x}$, $y_2 = xe^{2x}$, $W = e^{4x}$, set up the two integrals in the variation-of-parameters formula (don't evaluate).
-

A second forcing term with no elementary trial form

- 15 Solve $y'' - 2y' + y = \frac{e^x}{1+x^2}$ using variation of parameters ($y_1 = e^x$, $y_2 = xe^x$, $W = e^{2x}$).
-

A closing Wronskian and setup pair

- 16 For $y_1 = e^{-2x}$, $y_2 = e^{5x}$ (solutions of $y'' - 3y' - 10y = 0$), compute $W = y_1 y_2' - y_1' y_2$.
- 17 For $y'' + 9y = \csc 3x$, with $y_1 = \cos 3x$, $y_2 = \sin 3x$, $W = 3$, set up the two integrals in the variation-of-parameters formula (don't evaluate).
-

A final full worked example

- 18 Solve $y'' + 4y = \sec 2x \tan 2x$ using variation of parameters (continuing the setup from an earlier question, with $y_1 = \cos 2x$, $y_2 = \sin 2x$, $W = 2$).
-

A last Wronskian check

19 For $y_1 = x$, $y_2 = x^2$ (solutions of the equation $x^2 y'' - 2xy' + 2y = 0$ on $x > 0$), compute $W = y_1 y_2' - y_1' y_2$.